

```
# Load data
```

```
dat <- read.table(header = TRUE, text = '  
  Name  Subject Semester1 Semester2  
  Samma Maths    96         94  
  Sandy English   76         51  
  Devesh German   76         95  
  Rakesh Maths    50         63  
  Priya English   62         80  
  Kranti Maths    92         92  
  William German  87         75  
' )
```

```
# View data
```

```
print(dat)
```

```
# Transpose examples
```

```
res1 <- proc_transpose(dat) # No parameters
```

```
res2 <- proc_transpose(dat, name = VarName, id = Name) # With name and id
```

```
res3 <- proc_transpose(dat, id = v(Name, Subject)) # Two ID variables
```

```
res4 <- proc_transpose(dat, by = Name, id = Subject, name = Semester) # Grouped by Name
```

```
res5 <- proc_transpose(dat, by = Subject, id = Name) # Grouped by Subject
```

```
res6 <- proc_transpose(dat, by = v(Name, Subject)) # Two BY variables
```

```
# Sorting examples
```

```
sorted1 <- proc_sort(dat) # Default sort
```

```
# Note: 'Score' column does not exist
```

```
# sorted2 <- proc_sort(dat, by = Score)
```

```
# sorted3 <- proc_sort(dat, by = Score, order = descending)
```

```
sorted4 <- proc_sort(dat, by = Name, keep = v(Name, Semester1)) # Keep existing column
```

```
sorted5 <- proc_sort(dat, by = Subject)
```

```
# Data access examples
```

```
head(dat)
```

```
tail(dat)
```

```
subset(dat, Subject == "Maths") # Example subset
```

```
dat[3,]      # Row 3
```

```
dat[5:10,]   # Rows 5 to 10
```

```
dat[2:5, 3:4] # Rows 2 to 5, columns 3 and 4 (Semester1, Semester2)
```